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A randomised controlled trial of a digital intervention (Renewed) to support symptom management, wellbeing and quality of life in cancer survivors

Paul Little, Katherine Bradbury, Beth Stuart, Jane Barnett, Adele Krusche, Mary Steele, Elena Heber, Steph Easton, Kirsten Ailsa Smith, Joanna Slodkowska-Barabasz, Liz Payne, Teresa Corbett, Guiqing Yao, Sebastien Pollet, Jazzine Smith, Judith Joseph, Megan Lawrence, Dankmar Bohning, Tara Cheetham-Blake, Diana EcclesClaire Foster, Adam W A Geraghty, Geraldine Leydon, Andre Muller, Richard Neal, Richard Osborne, Shanaya Rathod, Alison Richardson, Chloe Grimmett, Geoff Sharman, Roger Bacon, Lesley Turner, Richard Stephens, Tamsin Burford, [Laura Wilde](#), Karen Middleton, Megan Liddiard, Kirsty Rogers, James Raftery, Shihua Zhu, Fran Webley, Gareth Griffiths, Jaqui Nutall, Trudie Chalder, Clare E Wilkinson, Eila Watson, Lucy Yardley

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Abstract

Background: Many cancer survivors following primary treatment have prolonged poor quality of life.

Aim: To determine the effectiveness of a bespoke digital intervention to support cancer survivors.

Design: Pragmatic parallel open randomised trial.

Setting: UK general practices.

Methods: People having finished primary treatment (≤ 10 years previously) for colo-rectal, breast or prostate cancers, with European-Organization-for-Research-and-Treatment-of-Cancer QLQ-C30 score <85 , were randomised by online software to: 1) detailed 'generic' digital NHS support ('LiveWell'; $n=906$), 2) a bespoke complex digital intervention ('Renewed'; $n=903$) addressing symptom management, physical activity, diet, weight loss, distress, or 3) 'Renewed-with-support' ($n=903$): 'Renewed' with additional brief email and telephone support.

Results: Mixed linear regression provided estimates of the differences between each intervention group and generic advice: at 6 months (primary time point: n 's respectively 806;749;705) all groups improved, with no significant between-group differences for EORTC QLQ-C30, but global health improved more in both intervention groups. By 12 months there were: small improvements in EORTC QLQ-C30 for Renewed-with-support (versus generic advice: 1.42, 95% CIs 0.33-2.51); both groups improved global health (12 months: renewed: 3.06, 1.39-4.74; renewed-with-support: 2.78, 1.08-4.48), dyspnoea, constipation, and enablement, and lower NHS costs (generic advice £265: in comparison respectively £141 (153-128) and £77 (90-65) lower); and for Renewed-with-support improvement in several other symptom subscales. No harms were identified.

Conclusion: Cancer survivors quality of life improved with detailed generic online support. Robustly developed bespoke digital support provides limited additional short term benefit, but additional longer term improvement in global health

enablement and symptom management, with substantially lower NHS costs.

Original language English

Article number BJGP.2023.0262

Pages (from-to) (In-Press)

Number of pages 22

Journal [British Journal of General Practice](#)

Volume (In-Press)

Early online date 20 Dec 2023

DOIs • <https://doi.org/10.3399/BJGP.2023.0262>

Publication status E-pub ahead of print - 20 Dec 2023

Bibliographical note

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Funder

This article presents independent research funded by the National Institute for Health Research (NIHR) under its Programme Grants for Applied Research programme (grant ref No RP-PG-0514-20001). LY is an NIHR Senior Investigator and her research programme is partly supported by NIHR Applied Research Collaboration (ARC)-West and NIHR Health Protection Research Unit (HPRU) for Behavioural Science and Evaluation. The Renewed intervention was developed using LifeGuide software, which is partly supported by the NIHR Southampton Biomedical Research Centre (BRC).

Keywords

- cancer survivors
- global health
- resource use

UN SDGs

This output contributes to the following UN [Sustainable Development Goals \(SDGs\)](#)

- SDG 3 - Good Health and Well-being

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Little, P., Bradbury, K., Stuart, B., Barnett, J., Krusche, A., Steele, M., Heber, E., Easton, S., Smith, K. A., Slodkowska-Barabasz, J., Payne, L., Corbett, T., Yao, G., Pollet, S., Smith, J., Joseph, J., Lawrence, M., Bohning, D., Cheetham-Blake, T., ... Yardley, L. (2023). [A randomised controlled trial of a digital intervention \(Renewed\) to support symptom management, wellbeing and quality of life in cancer survivors](#). *British Journal of General Practice*, (In-Press), (In-Press). Article BJGP.2023.0262. Advance online publication. <https://doi.org/10.3399/BJGP.2023.0262>
[A randomised controlled trial of a digital intervention \(Renewed\) to support symptom management, wellbeing and quality of life in cancer survivors](#). / Little, Paul; Bradbury, Katherine; Stuart, Beth et al. In: *British Journal of General Practice*, Vol. (In-Press), BJGP.2023.0262, 20.12.2023, p. (In-Press).

Research output: Contribution to journal › Article › peer-review

Little, P, Bradbury, K, Stuart, B, Barnett, J, Krusche, A, Steele, M, Heber, E, Easton, S, Smith, KA, Slodkowska-Barabasz, J, Payne, L, Corbett, T, Yao, G, Pollet, S, Smith, J, Joseph, J, Lawrence, M, Bohning, D, Cheetham-Blake, T, Eccles, D, Foster, C, Geraghty, AWA, Leydon, G, Muller, A, Neal, R, Osborne, R, Rathod, S, Richardson, A, Grimmett, C, Sharman, G, Bacon, R, Turner, L, Stephens, R, Burford, T, [Wilde, L](#), Middleton, K, Liddiard, M, Rogers, K, Raftery, J, Zhu, S, Webley, F, Griffiths, G, Nutall, J, Chalder, T, Wilkinson, CE, Watson, E & Yardley, L 2023, '[A randomised controlled trial of a digital intervention \(Renewed\) to support symptom management, wellbeing and](#)

quality of life in cancer survivors', [British Journal of General Practice](#), vol. (In-Press), BJGP.2023.0262, pp. (In-Press). <https://doi.org/10.3399/BJGP.2023.0262>

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Little, Paul ; Bradbury, Katherine ; Stuart, Beth et al. / [A randomised controlled trial of a digital intervention \(Renewed\) to support symptom management, wellbeing and quality of life in cancer survivors](#). In: [British Journal of General Practice](#). 2023 ; Vol. (In-Press). pp. (In-Press).

@article{ffb845c264dd45cb986e65dce15f16ff,

title = "A randomised controlled trial of a digital intervention (Renewed) to support symptom management, wellbeing and quality of life in cancer survivors",

abstract = "Background: Many cancer survivors following primary treatment have prolonged poor quality of life.Aim: To determine the effectiveness of a bespoke digital intervention to support cancer survivors.Design: Pragmatic parallel open randomised trial.Setting: UK general practices.Methods: People having finished primary treatment (<= 10 years previously) for colorectal, breast or prostate cancers, with European-Organization-for-Research-and-Treatment-of-Cancer QLQ-C30 score <85, were randomised by online software to: 1) detailed {\textquoteleft}generic{\textquoteright} digital NHS support ({\textquoteleft}LiveWell{\textquoteright};n=906), 2) a bespoke complex digital intervention ({\textquoteleft}Renewed{\textquoteright};n=903) addressing symptom management, physical activity, diet, weight loss, distress, or 3) {\textquoteleft}Renewed-with-support{\textquoteright} (n=903): {\textquoteleft}Renewed{\textquoteright} with additional brief email and telephone support. Results: Mixed linear regression provided estimates of the differences between each intervention group and generic advice: at 6 months (primary time point: n{\textquoteright}s respectively 806;749;705) all groups improved, with no significant between-group differences for EORTC QLQ-C30, but global health improved more in both intervention groups. By 12 months there were: small improvements in EORTC QLQ-C30 for Renewed-with-support (versus generic advice: 1.42, 95% CIs 0.33-2.51); both groups improved global health (12 months: renewed: 3.06, 1.39-4.74; renewed-with-support: 2.78, 1.08-4.48), dyspnoea, constipation, and enablement, and lower NHS costs (generic advice £265: in comparison respectively £141 (153-128) and £77 (90-65) lower); and for Renewed-with-support improvement in several other symptom subscales. No harms were identified.Conclusion: Cancer survivors quality of life improved with detailed generic online support. Robustly developed bespoke digital support provides limited additional short term benefit, but additional longer term improvement in global healthenablement and symptom management, with substantially lower NHS costs.",

keywords = "cancer survivors, global health, resource use",

author = "Paul Little and Katherine Bradbury and Beth Stuart and Jane Barnett and Adele Krusche and Mary Steele and Elena Heber and Steph Easton and Smith, {Kirsten Ailsa} and Joanna Slodkowska-Barabasz and Liz

Payne and Teresa Corbett and Guiqing Yao and Sebastien Pollet and Jazzine Smith and Judith Joseph and Megan Lawrence and Dankmar Bohning and Tara Cheetham-Blake and Diana Eccles and Claire Foster and Geraghty, {Adam W A} and Geraldine Leydon and Andre Muller and Richard Neal and Richard Osborne and Shanaya Rathod and Alison Richardson and Chloe Grimmett and Geoff Sharman and Roger Bacon and Lesley Turner and Richard Stephens and Tamsin Burford and Laura Wilde and Karen Middleton and Megan Liddiard and Kirsty Rogers and James Raftery and Shihua Zhu and Fran Webley and Gareth Griffiths and Jaqui Nutall and Trudie Chalder and Wilkinson, {Clare E} and Eila Watson and Lucy Yardley",

note = "This is an Open Access article distributed under the terms of the Creative Commons Attribution 4.0 License (<http://creativecommons.org/licenses/by/4.0/>).",

year = "2023",

month = dec,

day = "20",

doi = "10.3399/BJGP.2023.0262",

language = "English",

volume = "(In-Press)",

pages = "(In--Press)",

journal = "British Journal of General Practice",

issn = "0960-1643",

publisher = "Royal College of General Practitioners",

}

TY - JOUR

T1 - A randomised controlled trial of a digital intervention (Renewed) to support symptom management, wellbeing and quality of life in cancer survivors

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PY - 2023/12/20

Y1 - 2023/12/20

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KW - cancer survivors

KW - global health

KW - resource use

U2 - 10.3399/BJGP.2023.0262

DO - 10.3399/BJGP.2023.0262

M3 - Article

C2 - 38164562

SN - 0960-1643

VL - (In-Press)

SP - (In-Press)

JO - British Journal of General Practice

JF - British Journal of General Practice

M1 - BJGP.2023.0262

ER -



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